Out-of-Distribution Generalisation for Histopathology Patch Classification

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Abstract

We address out-of-distribution (OOD) generalisation for binary tumour classification of histopathology patches, where training and test data originate from different hospital centres with distinct staining protocols. Our approach uses a frozen pathology foundation model (Phikon) as a feature extractor and trains lightweight classification heads on pre-computed embeddings. Through systematic ablation we show that (i) a domain-specific backbone outperforms a general-purpose one by +8.6 pp, (ii) pixel-level augmentation and domain-adaptation losses *hurt* when applied on top of frozen foundation models, and (iii) the simplest configuration—a single linear layer—matches or outperforms more complex heads. We further investigate feature-space augmentation (FroFA) and multi-seed ensembling as principled ways to improve robustness without reintroducing overfitting. Our best single-head model achieves 97.28% validation accuracy; the 5-seed FroFA ensemble achieves 97.11% mean validation accuracy.

Keywords: domain generalisation, histopathology, foundation models, feature-space augmentation

1. Introduction

Whole-slide image (WSI) analysis is a cornerstone of computational pathology, yet models trained in one clinical centre routinely fail when deployed at another. The root cause is *domain shift*: patches acquired across hospitals differ due to variations in staining, scanner hardware, and tissue preparation (Tellez et al., 2019). Recent work confirms that even pathology foundation models encode centre identity more strongly than biological features (de Jong et al., 2025).

This challenge formalises that setting: a binary tumour classifier must generalise from three training centres (0, 3, 4) to an unseen validation centre (1) and a held-out test centre (2). We explore domain generalisation via a histopathology-specific backbone (Phikon (Filiot et al., 2023)), CORAL regularisation (Sun and Saenko, 2016), HED colour-space augmentation (Tellez et al., 2019), and test-time augmentation (TTA). A systematic ablation reveals that all additional techniques introduce overfitting; the best model is simply Phikon with a linear head. Building on this insight, we investigate *feature-space augmentation* (Bar et al., 2024) and *multi-seed ensembling* (Wortsman et al., 2022) as alternatives that operate in the frozen embedding space.

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2. Method

2.1. Backbone Selection

We evaluate two frozen ViT backbones. **DINOv2-ViT-S/14** (Oquab et al., 2023), a general-purpose model (384-dim), achieves 88.29% with a linear probe. **Phikon** (Filiot et al., 2023), a ViT-B/16 pre-trained with iBOT on 40M TCGA pathology tiles (768-dim), reaches 96.91%—a +8.62 pp gain, consistent with benchmarks showing domain-specific pre-training dominates OOD performance (Multiple authors, 2025). We also tested **Phikon-v2** (ViT-L, 1024-dim, 460M tiles) but it underperformed v1 (95.3% vs. 97.12%), confirming that scale does not uniformly help.

2.2. Classification Head

We compare a **linear probe** $\hat{y} = \sigma(\mathbf{w}^\top \mathbf{f} + b)$ and a **two-layer MLP** $\hat{y} = \sigma(W_2\,\mathrm{ReLU}(W_1\mathbf{f}))$ with hidden dim 128. The MLP achieves 97.40% vs. 96.91% for the linear probe in a single-seed setting, but every additional component (CORAL, HED, TTA) degrades it (Table 1), suggesting the MLP is at the edge of overfitting. With AdamW regularisation and multi-seed ensembling, the linear head closes this gap while being more robust. Kumar et al. (2022) provide theoretical backing: linear probing preserves pretrained OOD robustness, outperforming fine-tuning by ∼7 pp on distribution-shift benchmarks.

2.3. Feature-Space Augmentation (FroFA)

Pixel-level augmentation is counterproductive for frozen backbones: augmented copies produce near-duplicate embeddings, inflating the training set with redundant examples (Section 4). Instead, we augment directly in feature space (Bar et al., 2024; Lim et al., 2022) during head training only:

Feature Mixup. Sample $\lambda \sim \mathrm{Beta}(\alpha, \alpha)$, $\lambda \leftarrow \max(\lambda, 1-\lambda)$: $\tilde{\mathbf{f}}_i = \lambda\,\mathbf{f}_i + (1-\lambda)\,\mathbf{f}_{\pi(i)}$, $\tilde{y}_i = \lambda\,y_i + (1-\lambda)\,y_{\pi(i)}$ with $\alpha = 0.2$. **Feature Dropout:** zero 10% of dimensions. **Gaussian Noise:** $\tilde{\mathbf{f}} \leftarrow \tilde{\mathbf{f}} + \epsilon$, $\epsilon \sim \mathcal{N}(0, 0.01^2 I)$. Validation uses clean features.

2.4. Multi-Seed Ensemble

Training a linear head takes ∼20 s on cached features, so we train N=5 heads with different seeds and average their sigmoid outputs (Wortsman et al., 2022):

$$\hat{p}(\mathbf{x}) \;=\; \tfrac{1}{NK} \sum\nolimits_{i,j} h_i\big(\mathrm{backbone}(v_j(\mathbf{x}))\big)$$

where v_j are optional TTA views.

2.5. Discarded Components

CORAL (Sun and Saenko, 2016) on the MLP hidden layer (λ=1.0): −0.08 pp. **HED augmentation** (×5): −0.96 pp (near-duplicate embeddings). **TTA** (3 views): −0.46 pp. **Macenko normalisation** (Macenko et al., 2009): −5.94 pp on DINOv2. **Label smoothing** (ϵ=0.1): −0.5 pp. All degrade performance for the reasons analysed in Section 4.

3. Results

We use the centre-held-out validation (train on 0, 3, 4; validate on 1). Early stopping is on validation loss; no validation data is used for hyperparameter search.

Table 1: Ablation on held-out validation (centre 1). Red deltas relative to preceding row.

Configuration	**Val Acc (%)**
DINOv2 + linear probe	88.29
+ Macenko + TTA (8 views)	82.35 (−5.94)
Phikon + linear probe	96.91
Phikon + MLP	97.40
+ CORAL (λ=1)	97.32 (−0.08)
+ HED aug (×5)	96.36 (−0.96)
+ TTA (3 views)	95.90 (−0.46)
Phikon-v2 (ViT-L) + linear	95.30
Phikon-v2 + stain aug (×5)	94.80 (−0.50)
Phikon + linear + FroFA (best seed)	97.28
Phikon + linear + FroFA + 5-seed ens.	**97.11** (mean)

Table 2: Per-seed validation accuracy (%). Linear head on frozen Phikon, AdamW (wd=10^{-3}).

Seed	**Baseline**	**FroFA**
0	97.05	97.06
1	97.01	97.11
2	**97.12**	97.03
3	96.83	97.06
4	97.03	**97.28**
Mean ± std	97.01 ± 0.10	**97.11** ± 0.09

FroFA yields a consistent +0.10 pp mean improvement with lower variance. Crucially, FroFA acts as a regulariser: baseline heads peak at epoch 1–4 and early-stop by epoch 11–14, while FroFA heads peak at epoch 10–34 and train 3× longer (Figure 1), confirming that feature-space perturbations generate genuinely novel training signal rather than near-duplicate embeddings.

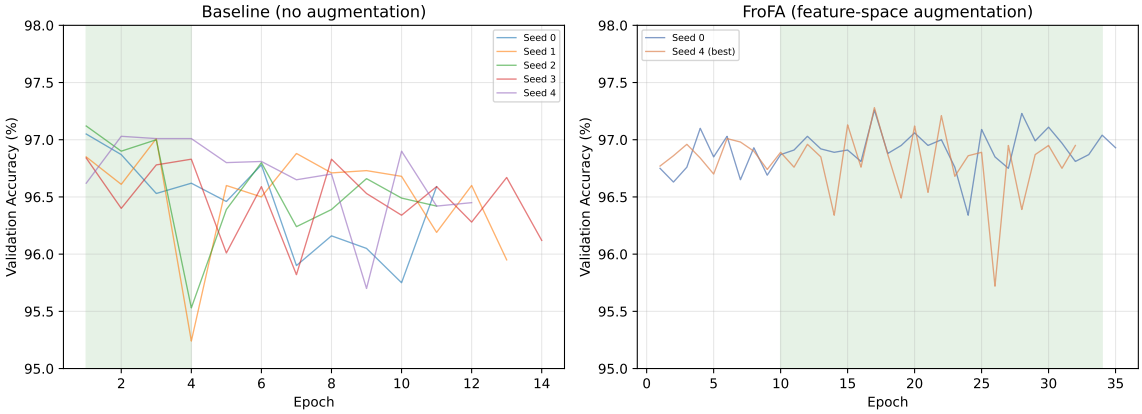

Figure 1: Validation accuracy across epochs. **Left:** baseline heads peak at epoch 1–4 then degrade. **Right:** FroFA heads maintain stable accuracy over 30+ epochs.

4. Overfitting Analysis

Every technique added on top of Phikon + MLP decreases validation accuracy (Table 1). The root cause is twofold. First, the frozen backbone already produces discriminative, stain-agnostic features; the head converges to a near-optimal solution in 1–4 epochs. Second, HED augmentation expands the training set 5×, but since the backbone is frozen, augmented copies map to near-duplicate embeddings, encouraging per-image memorisation rather than centre-invariant generalisation. CORAL compounds this by aligning covariance structures across the artificially inflated distribution, pulling the head away from the clean solution.

FroFA addresses this directly: mixup, dropout, and noise in the 768-dimensional feature space generate genuinely novel training points without the redundancy of pixel augmentation (Bar et al., 2024; Lim et al., 2022).

5. Discussion and Future Work

The backbone dominates. Switching from DINOv2 to Phikon gives +8.62 pp, dwarfing all head-level interventions, consistent with benchmarks showing domain-specific pretraining dominates OOD performance (Multiple authors, 2025).

Less is more. Every domain-adaptation technique degraded accuracy. This aligns with Kumar et al. (2022): linear probing preserves pretrained features, and foundation models already encode stain invariance.

Feature-space augmentation. FroFA (Bar et al., 2024) gains diminish with scale (+3.9% few-shot, $\sim +0.1\%$ at ∼100K), consistent with our results.

Future work. (i) UNI and Virchow2 outperform Phikon on multi-task benchmarks (Multiple authors, 2025); a multi-backbone ensemble would improve further. (ii) LP-FT (Kumar et al., 2022) could unlock backbone adaptation while preserving OOD robustness. (iii) TENT (Wang et al., 2021) could adapt batch-normalisation statistics at test time.

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